

Securing Shared University Lab Computers

Project Brief: The Problem, the Solution, and How to Apply It

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1. The Problem (in one sentence)

On shared university lab computers, the default settings of Chrome/Edge and Windows allow one student's saved passwords, synced personal data, and session to leak to the next person who uses that machine — with no malware, no hacking, just default behaviour.

2. What Actually Happens (the issues)

#	Issue	What happens
1	Leftover profile	A student signs into Chrome, saves passwords, leaves. The next student opens chrome://settings/passwords, types the s
2	Sync bleed	With sync on, Chrome downloads the student's entire personal life onto the lab PC — passwords, autofill, histor
3	Leftover files	Even after sign-out, the profile folder stays on disk. A free tool (ChromePass, LaZagne) reads the saved passwords off
4	Keyloggers	A student installs a hidden Python keylogger or a malicious browser extension. It captures every keystroke of everyone
5	Hardware keylogger	A physical USB device placed between the keyboard and PC records everything. Software cannot detect it.

*Real-world cost: this is how a UIU student's credentials were stolen and her images leaked, and how exam questions get leaked.
Real harm — especially to women — from a setting nobody changed.*

3. Our Solution (the principle)

Do not train users to be careful — remove the opportunity for harm. Take the juice off the table. Every fix is a system/admin change, not a user instruction. All controls are free, built into Windows/Chrome or open-source. The solution has two halves: prevention (stop the leak from being possible) and detection (see anything that slips through).

4. How We Apply It (the controls)

The backbone — four controls that close most of the problem:

- **Standard user accounts** — students are not local admins. Removes the power to install keyloggers. (Issue 4)
- **One reset mechanism** — Windows Shared PC Mode (delete profile on logout) OR Unified Write Filter (wipe to clean state on reboot). Pick one. Destroys all leftover data every cycle. (Issues 1, 2, 3)
- **Browser lockdown via Group Policy** — BrowserSignin = 0 (can't sign in), SyncDisabled = 1 (no sync), PasswordManagerEnabled = 0 (nothing saved), ForceEphemeralProfiles (fresh profile each time), ExtensionInstallBlocklist = ["*"] + small allowlist (no malicious extensions). (Issues 1, 2, 3, 4)
- **Detection: Sysmon + Wazuh** — logs and alerts on credential-theft attempts, hidden keyloggers, and suspicious data exfiltration. Catches what prevention misses. (Issue 4)

The supporting controls:

- **AppLocker + PowerShell Constrained Language Mode** — blocks Python scripts and unapproved executables from running. (Issue 4)
- **Defender ASR rule for LSASS** — blocks credential dumping (defence-in-depth).
- **USB device control + physical inspection + on-screen keyboard for sensitive logins** — the only defence against hardware keyloggers. Honest: reduces, does not eliminate. (Issue 5)

5. How We Prove It Works

Lab A = apply the controls. Lab B = leave unchanged (control group). Run the extraction test on both before; apply controls to Lab A over a 30-day window; run the same test after. Lab A should leak nothing; Lab B unchanged. Add an anonymous before/after survey on student awareness and behaviour.

Metric	Lab A Before	Lab A After	Lab B (control)
Credential extraction success rate	high (measure)	near zero (target)	unchanged

Machines with sync/sign-in disabled	measure	100%	unchanged
Machines resetting state each cycle	measure	100%	unchanged
Survey: % unaware of the risk	measure	measure	n/a

The headline result is the number, e.g. "Lab A went from X of 15 machines leaking to 0 of 15."

6. The Three Tools You Build

Tool	What it does	Honest claim
Audit script	Checks a machine's current settings and reports what is exposed	Measures exposure; fixes nothing
Disposable-profile launcher	Launches Chrome in a temp folder, sync off, auto-deleted on close	Kills sync/profile leak only; does NOT stop keyloggers
Extraction-test harness	Seeds a test account, attempts extraction, records pass/fail	The measurement engine; before and after

7. The Honest Line You Never Cross

- Software cannot stop hardware keyloggers. Say so.
- The launcher does not stop keyloggers — only admin-side controls (standard user + AppLocker) do. Say so.
- Use only your own test accounts. Never touch real student data.
- Get written permission before touching any machine. The Cyber Security Ordinance 2025 applies.

Problem → what happens → solution → how to apply → how to prove → what to build → where to stay honest. Start with the audit script — it is the simplest tool and the first thing to demo. Everything else builds from there.

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